

A Comparative Study of Text-Based Emotion Detection Techniques for Emotion Recognition on Social Media Data

Diksha Shukla
Dept. of Computer Science
Babasaheb Bhimrao Ambedkar
University
Lucknow, India
dikshashukla73@gmail.com

Prof. Sanjay K. Dwivedi
Dept. of Computer Science
Babasaheb Bhimrao Ambedkar
University
Lucknow, India
skd200@yahoo.com

Abstract— Nowadays, with the rise of the internet social media have emerged as a new platform that provides a large amount of user-generated content that can be analyzed or mined to detect the emotions of the users. However, advancement in technology allows users to share their views through audio/video, speech, and facial expressions but people mostly use text for communication over social platforms. Hence it is required to automatically recognize emotions from these posts and provide responses to users in today's online world. Emotion Detection from text is a popular topic in Natural Language Processing (NLP) due to its vast applications in human-computer interaction, artificial intelligence, marketing, etc. Researchers from different domains have proposed algorithms for emotion detection from the text; still, no method gave a complete solution. Hence, this paper aims to explore several emotion detection techniques such as machine learning (ML), Deep learning (DL), and hybrid ensemble techniques to detect emotions from social media conversations. Then these different classifiers are analyzed, tested, and compared with each other based on different parameters namely F1-score, recall, precision, accuracy, and confusion matrix. The classifier that gets the highest score among all other classifiers is termed as best text-based emotion classifier.

Keywords—*Emotion Detection(ED), Natural Language Processing, Machine Learning, Deep Learning*

I. INTRODUCTION

Emotions are an important aspect in the human communication and have many applications in the human computer interaction and in information retrieval [Chetan R.Chopade, 2013]. Sentiment analysis (SA) focuses on studying and identifying sentiments in order to find out the attitude or opinion of a people on particular context and classifying them as positive, negative and neutral. ED is a branch of SA that identifying emotions from particular context and then analyze it using some categorical / dimensional class models. There are so many areas/applications in which text based emotion detection & sentiments have an important aspects including observing mental health of people, making efficient E-learning system, detecting public opinion on any political and social event, improving business strategies according to customer reviews or opinion and so on. Analyzing sentiment from text is

comparatively less complex than identifying emotions from text due to the versatility& ambiguity of human emotions[R Murthy et al., 2021].

Recognizing emotional state of a person from a text is quite challenging as different emotions sometimes have same expression, emotions are not explicitly defined, sometimes one paragraph contain different emotions with different polarities and therefore difficult to detect [Shivhare et al., 2012]. Emotions may also vary based on location, gender, personality, culture and many more psychological parameters. Researchers from different domain have proposed so many algorithms/approaches for emotion detection such as: machine learning, deep learning, rule based approach, and hybrid techniques. At first, they introduced keyword based approach. Later on, learning based models are come into existence. Different supervised/unsupervised machine learning models are used for emotion detection. However, these models give better result for small data, for large scale data, deep learning models show better accuracy. In the search of high accuracy, hybrid models are used by combing different techniques/approaches. [Santosh Kumar Bharti et al., 2022].

Recognizing emotions from text is a popular topic in recent years. Still there is no method that gave complete solution for emotion detection from text. This paper studies various methods such as ML, DL and hybrid ensemble technique for emotion classification. Then these different classifiers are analyzed, tested and compared with each other based on different parameters in order to find out the classifier that performed well as compared to other classifiers for emotion detection.

The remaining section of the article is organized as follows: Section II. discusses the related work, Section III . discusses the emotion detection approaches, Section IV. defines the methodology, Section V. contains the result & discussion and finally section VI. discusses the conclusion.

II. RELATED WORK

ED from text is a popular topic. In this section, some of the existing work and their contribution to emotion detection are listed.

Jonathan and his colleague [Jonathan et al., 2017], explores various algorithms such as NB, Fast-Large Margin, Generalized linear model, DT, SVM, RF and ANN to recognize emotions from social media posts. Affective tweets dataset which consists of more than 7000 emotions labelled utterances are used. Their result shows that generalized linear model provides best results as compared to other methods and DT & RF were not suitable for this problem.

[Samar Fathy et al., 2017], shows that accuracy rate can be improved by using proper pre-processing and feature extraction (FE) technique. To remove noise from the dataset they undergoes with preprocessing steps. For feature extraction they used two methods, Bow and count Vectorizer. For evaluation, they used hotel, car and mobile dataset and Bert algorithm is used for emotion classification.

[Chetan R.Chopade, 2013], presented a literature review of text based emotion recognition. They describe the different ED methods, and also address the text normalization for handling both plain text and short messaging language.

Sasidhar and his colleagues [T Tulasi et al., 2020], first created a Hindi-English code mixed text based dataset and then annotate the dataset using basic emotions-anger, happy and sad. For classification, deep neural network models are used and to generate feature vectors pretrained bilingual model is used. Their experimental result shows that CNN-BiLSTM gave better result with 83.21% accuracy as compared to other models.

Nilesh Shelke and his colleagues [Nilesh Shelke et al., 2022], addressed the irrelevant feature extraction issues. For this they proposed a LRA-DNN. For evaluation publically available datasets are used. Their results shows that proposed leaky-relu activated deep neural network method gave maximum accuracy, specificity and sensitivity 94.77%, 95.91% and 92.23% respectively as compared to other ANN, DNN and CNN methods.

[Ankush Chatterjee et al., 2019] have proposed a novel Deep learning based approach called Sentiment and Semantic based Emotion Detector (SS-BED) to detect emotions-sad, angry and happy from dialogues. They used both sentiment and semantic features for ED. Their results show that proposed approach outperforms other ML models as well as off-the-shelf DL models.

Parvin and his colleague [Tanzia Parvin et al., 2021], investigates several ML algorithms- LR, SVM, MNB, KNN, RF, DT and a combination of SVM, RF, LR (ensemble method) and Adaptive Boosting (AdaBoost) with tf-idf and Bag- of- words FE techniques for emotion detection from Bengali text. Their result shows that the highest weighted f1-score 62.39% was achieved by hybrid ensemble method with tf-idf as compared to other methods.

[Douji yasmina et al., 2016], have presented an unsupervised ML algorithm based emotion detection system. They used comments from YouTube as dataset. Proposed

system gives an average accuracy of 68.82% and average precision of 92.75%.

[Jonathan Herzig et al., 2017], have Proposed a new method based on pre-trained dense word embedding representation. They proposed an ensemble approach combining both dense and sparse representation. For ED they used 5 datasets from various domains and their experimental result shows that an ensemble approach that combines embedding representation and BoW outperforms for domain specific dataset. Proposed approach requires little computing power as compared to other deep learning methods.

[Santosh et al., 2022], have proposed a model which is a combination of DL and ML techniques (Hybrid model) to identify emotions from text. SVM is used as a ML approach and CNN & Bi-GRU were used as DL techniques. For evaluation, 3 datasets namely WASSA, Emotion-Stimulus and the ISEAR are used. The advantages of the proposed approach are: it can work on tweets, keywords, lexicon dialogs and multitext sentences. Their experimental result shows that highest accuracy of 78.97% was achieved by SVM, Bi-GRU model gives an accuracy of 79.46% and F1-score achieved by CNN model is 80.76%. The proposed hybrid approach achieved an accuracy of 80.11%.

III. EMOTION DETECTION APPROACHES

Based on available literature, different approaches exist for text-based ED such as: keyword-based approach, Lexicon-Affinity Approach, ML based approach, DL based approach, and Hybrid method.

A. Keyword-Based Approach

This method is easy to implement as compared to other methods. In keyword based approach we simply find out the keywords which are similar to some predefined emotional words and then match them with these words. First we find out the words that represent emotions in the given sentence. Then these words that we are found in the previous step are matched with a number of words that represent emotions according to some particular emotion models. After that the emotions which matches with the word is considered as the emotion of the word/sentence. These words are classified as happy, angry, fear, surprise, disgust etc [chopade 2013]. WorldNet, DepecheMood, and SentiWordNet dictionaries are used as a keyword dictionary.

B. Lexicon-Affinity Approach

In this method, lexical affinity assigned probabilistic affinities to random emotional words apart from the identification of keywords as in keywords based approach. These probabilities are a part of linguistic corpora. For example, a probabilistic affinity assigned as “positive” for a word ‘happy’ and a probabilistic affinity assigned as “negative” for a word ‘sad’ [Acheampong et al., 2019, Shrivhare et al., 2012].

C. Learning-Based Method

Machine learning methods classifying input text into different emotion categories by using a pre-trained classifier. Various supervised and unsupervised ML techniques are used to build a model which is designed to train the classifier and then test the classifier using test data. Supervised techniques achieve better accuracy than unsupervised techniques.

D. Hybrid Method

Hybrid method is the combination of keywords based method and learning based method or deep learning method in order to provide more accurate results. Many researchers combine any two previously defined methods to achieve the benefit of multiple methods for emotion classification.

E. Deep Learning Method

In the initial years, many supervised ML techniques such as SVM, NB and so on are used for text based emotion detection. However, these techniques do not extract the intrinsic features from the text explicitly and are not robust too. But through the years DL Models are being adopted by various researchers for emotion analysis as these techniques are more robust and with the help of deep layers they can extract the semantic features from the text explicitly [Acheampong et al., 2020].

TABLE I COMPARISON OF VARIOUS APPROACHES

Approach	Advantages	Disadvantages
Keyword Based	Labeled corpus is not required. Easy to implement.	Ambiguity in keyword definition Means multiple meaning of keywords. Ignoring linguistic information poses difficulty. Difficulty in recognizing sentences without Emotional keywords.
Lexicon-Affinity	No need for keyword identification.	Multiple categories of emotions are not represented. Lead to inaccuracies in emotion classification.
Learning Based	Dictionary is not necessary.	Annotation process becomes time consuming. Large labeled corpus is required Semantic information's are not extracted explicitly from the text.
Hybrid Method	Yields higher accuracy. Cost balancing in terms of information retrieval and risk minimization.	Difficulty in choosing the technique to enhance performance.
Deep Learning	Explicitly extract the semantic information from the text. More Robust.	Complex.

IV. METHODOLOGY

This section discusses about the proposed methodology of emotion detection for twitter messages.

Data collection played a vital role in any text classification task. An Emotion Dataset extracted from Kaggle is considered for Emotion Detection.

An Emotion Dataset for Emotion Recognition task is publically available dataset from Kaggle.Com. The dataset consists of 20000 English twitter messages crawled from twitter API. The column 'label' represents the six basic emotions (i.e. anger, love, joy, fear, surprise, and sadness). Table II. Illustrates the dataset profile. Each label has imbalanced number of value counts; with the surprise has the smallest number (i.e. 719), and joy has the largest number (i.e. 6761). So we performed down- scaling using NearMiss to the data in order to make the class balance. After sampling each and every class has a balanced number of value counts (719).

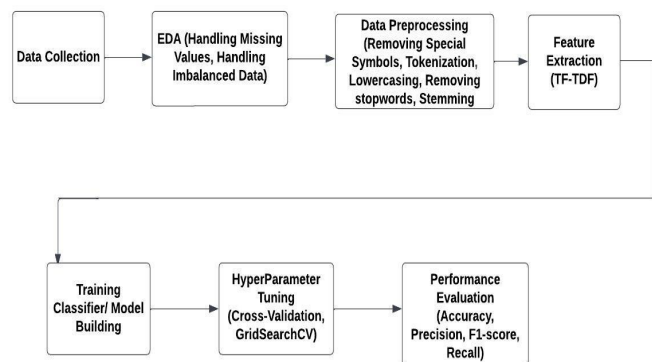


Fig. 1 Overview of Proposed System

After that the data was preprocessed to remove noise and unwanted textual information using various NLP preprocessing steps such as removing special characters, lower case, stop words removal, tokenization, stemming. NLTK tool is used for preprocessing. In the first step irrelevant and non-meaningful text was removed. After that, tokenization is done to convert sentences into smaller words. Then stopwords (e.g. I, the, in etc) were also removed and convert all the words in lower case. Finally, stemming process was done to extract the base of the words. Before passing the pre-processed dataset to various classifier/models, we used TF-IDF (term frequency-inverse document frequency) vectorization to extract the features from the preprocessed dataset. Now the next step is to train the model, several ML, DL and hybrid ensemble techniques were explored to train the Emotion Recognition dataset. In the training process, we used cross validation score (cv=5) and also explored various combinations of hyperparameters to enhance the results except for Deep Neural Network. Table IX. Represents the Deep neural network hyper-parameters. In Bidirectional LSTM, first word2Vec is used to extract feature vectors from the preprocessed data. The architecture of bidirectional LSTM consists of four layers. The i/p layer is the first layer which accept embedding feature vector of dimension 200 as input, the next two bidirectional LSTM layer with 100, 200 units in each layer. The 4th layer is output layer with 6

softmax units. A dropout of 0.2 is performed in all the layers. The architecture of BERT model consists of four layers, first layer is embedding layer which accepts input_ids and input_mask of length 70, the next two dense layer consists of 128, 32 units, The fourth layer is output layer with 6 softmax units, there is no dropout layer. Next model is hybrid ensemble technique which is the combination of SVM, logistic regression and Multinomial NB. After that each models was tested using the test dataset. In the next section the overall results are discussed.

TABLE II DATASET PROFILE

Class	Actual Data	Sampled Data
Sadness	5797	719
Joy	6761	719
anger	2709	719
Fear	2373	719
Love	1641	719
Surprise	719	719
Total	20000	4314

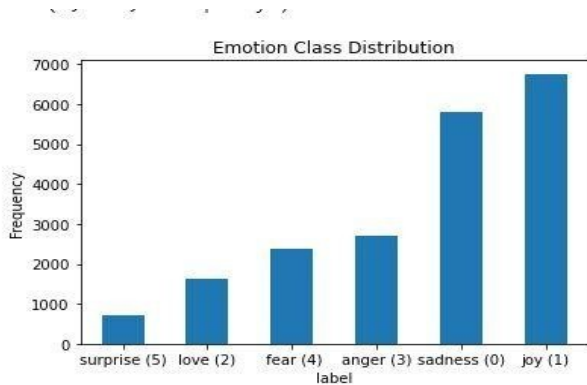


Fig. 2 Emotion Class Distribution

V. RESULTS & DISCUSSION

Several machine learning and deep learning models and hybrid ensemble technique namely SVM, Multinomial NB, KNN, Random Forest, Decision Tree, Logistic Regression, BiLSTM, BERT were explored in this research. In the training phase, 80% of the dataset are used. 20% dataset are used in the testing phase. During the training process, several hyper parameter combinations were also explored for SVM, NB, RF, LR, DT, KNN, Bi-LSTM and BERT algorithms. The hyperparameters that we have selected during the training process were recommended in this problem only; those hyperparameters may be changed for other problem domains. After Hyperparameter tuning the best parameters selected for SVM model are: ('C': 100, 'kernel': rbf, gamma: 0.005), as shown in table IV. In SVM model, the best F1 score achieved by love (i.e. 0.88), while the lowest F1 score achieved by fear (i.e. 0.74). SVM model achieved 0.81 accuracy score. In logistic regression model, we explored various combinations of hyperparameters such as 'penalty' value, 'C' value and 'multi- class' value as shown in table. III. And the best parameters selected for LR are ('C': 10, 'multi-class': OVR, 'penalty': l2). In this model, the best F1 score achieved by anger (i.e. 0.89), while the lowest F1 score achieved by surprise and fear (i.e. 0.77). LR model achieved 0.84 accuracy score. In Multinomial NB model five hyper-

parameters setting of alpha parameter (alpha) were explored (as shown in table.V). The parameter value alpha 1.0 gives the better result. The best F1 score achieved by Love (i.e. 0.83), while the lowest F1 score achieved by joy (i.e. 0.67). Multinomial NB model achieved 0.74 accuracy score.

K-Nearest Neighbor model also explores various combinations of hyperparameters (see table. VI). the best parameters selected for. K-Nearest Neighbor model are: (n_neighbors=15, algorithm=auto, weights=distance, p=2). The best F1 score achieved by Love (i.e. 0.81), while lowest F1 score achieved by surprise (i.e. 0.64). KNN achieved an accuracy score 0.70. RrandomForest model also explores four hyper-parameters setting of number of trees (n_estimators) and criterion (see table.VIII). The optimal parameters selected for random forest classifier are: (n_estimators=100, criterion=gini). The best F1 score achieved by love (i.e. 0.85), while the lowest F1 score achieved by joy (i.e. 0.64). RF model achieved an accuracy score 0.74. In the next training phase, the best parameters selected for Decision Tree model are: (criterion=gini, max_features=auto, Max_Depth=500, splitter=random) as shown in table VII. The best F1 score achieved by anger (0.69), while the lowest F1 score achieved by joy (i.e. 0.48). This model achieved 0.60 accuracy score. In the next training phase, hybrid ensemble technique is used which is a combination of SVM, LR, Multinomial NB, in this technique Love achieved the best F1 score (0.88) and surprise achieved the lowest F1 score (0.75). The model achieved the accuracy score of 0.82.

TABLE III HYPERPARAMETERS FOR LOGISTIC REGRESSION

C_Value	Multiclass	Penalty
10	OVR	L1
50	OVR	L2
100	OVR	Elasticnet
150	OVR	
200	OVR	
500	OVR	

TABLE IV HYPERPARAMETERS FOR SVM

C_Value	Kernel	Gamma
10	Linear	0.001
50	Poly	0.005
100	RBF	0.1
1000	Sigmoid	0.5
		0.8
		0.9

TABLE V HYPERPARAMETERS FOR MULTINOMIAL NB

alpha
0.000000000001
0.000000000001
0.00000000001
0.000001
0.001
0.01
0.1
1.0

In the next training phase, Bi- LSTM, BERT model are used. In Bi-LSTM, The best F1 score achieved by sadness

(i.e. 0.97), while the lowest F1 score achieved by surprise (i.e. 0.90). BiLSTM model achieved an accuracy score 0.94. In BERT, The best F1 score achieved by sadness (i.e. 0.97), while the lowest F1 score achieved by surprise (i.e. 0.83). BERT model achieved an accuracy score 0.94. The results are described in table X.

TABLE VI HYPERPARAMETERS FOR KNN

Number of Neighbors	weights	Algorithm	P
3	Uniform	Auto	1
5	Distance	Ball_tree	2
10	OVR	Kd_tree	
15	OVR	Brute	

TABLE VII HYPERPARAMETERS FOR DECISION TREE

Criterion	Splitter	Max_Features	Mmax_Depth
Gini	Best	Auto	50
Entropy	Random	sqrt	100
Log_Loss		Log2	150
			200
			250
			300
			500

TABLE VIII HYPERPARAMETERS FOR RANDOM FOREST

Num_of_Trees	Criterion
10	Gini
20	Entropy
30	Log_Loss
40	

TABLE IX HYPERPARAMETERS FOR NEURAL NETWORK

Algorithms	Hyperparameters
Bidirectional LSTM	{ input: embedding vector of dimension-200, 2 bidirectional LSTM layer: units-100, 200, Output:units-6, activation-softmax, dropout: 0.2 performed in each layer}
BERT	{Optimizer: Adam, learning_rate=1e-04, epsilon=1e-08, decay=0.01, no dropout}

TABLE X RESULTS

Algorithms	Class	Recall	Precision	F1
SVM	sadness	0.82	0.86	0.84
	Joy	0.84	0.79	0.81
	Love	0.91	0.86	0.89
	Anger	0.83	0.89	0.86
	Fear	0.78	0.70	0.74
	Surprise	0.70	0.76	0.73
Logistic Regression	sadness	0.84	0.90	0.87
	Joy	0.85	0.84	0.84
	Love	0.89	0.86	0.88
	Anger	0.89	0.89	0.89
	Fear	0.81	0.73	0.77
	Surprise	0.75	0.80	0.77
Multinomial NB	sadness	0.73	0.75	0.74
	Joy	0.64	0.71	0.67
	Love	0.82	0.84	0.83
	Anger	0.80	0.77	0.79
	Fear	0.77	0.64	0.70
	surprise	0.70	0.73	0.71
K-nearest Neighbors	sadness	0.65	0.66	0.65
	Joy	0.74	0.57	0.65
	Love	0.76	0.86	0.81
	Anger	0.76	0.85	0.80

Random Forest	Fear	0.69	0.70	0.69
	surprise	0.64	0.65	0.64
	sadness	0.69	0.68	0.68
	Joy	0.62	0.67	0.64
	Love	0.89	0.81	0.85
	Anger	0.78	0.86	0.82
Decision Tree	Fear	0.76	0.64	0.69
	Surprise	0.74	0.82	0.78
	Sadness	0.47	0.59	0.52
	Joy	0.55	0.43	0.48
	Love	0.63	0.70	0.66
	Anger	0.64	0.74	0.69
Hybrid Ensemble	Fear	0.63	0.53	0.58
	Surprise	0.68	0.68	0.68
	Sadness	0.83	0.88	0.85
	Joy	0.83	0.81	0.82
	Love	0.89	0.87	0.88
	Anger	0.86	0.88	0.87
Bi-LSTM	Fear	0.81	0.71	0.76
	Surprise	0.72	0.78	0.75
	Sadness	0.97	0.96	0.97
	Joy	0.96	0.96	0.96
	Love	0.93	0.89	0.91
	Anger	0.92	0.96	0.94
BERT	Fear	0.92	0.92	0.92
	Surprise	0.91	0.89	0.90
	Sadness	0.97	0.97	0.97
	Joy	0.94	0.95	0.95
	Love	0.86	0.83	0.84
	Anger	0.92	0.93	0.92
	Fear	0.90	0.91	0.91
	Surprise	0.84	0.81	0.83

VI. CONCLUSION

The main objective of this research is to explore various emotion recognition techniques in order to come up with emotion analysis that can effectively identify emotions from text. Algorithms are: Multinomial Naïve Bayes (MNB), Support vector machine (SVM), K- Nearest Neighbors (KNN), Decision tree (DT), Logistic Regression (LR), Random Forest (RF), Bidirectional LSTM (BiLSTM) and Bidirectional Encoder Representations from Transformers (BERT). In the training phase various hyperparameters are used to optimize the result. Emotion dataset extracted from Kaggle consists of 20000 labelled twitter text messages are used in this research. Our experimental result shows that Deep learning algorithm such as Bi-directional LSTM and BERT achieved the best accuracy rate (94%) as compared to other learning based models. In contrast, machine learning algorithm, Logistic regression achieved the best accuracy rate of 84%. Hybrid ensemble technique achieved the second best performance with an accuracy of 82%. MNB, KNN, RF and DT were not recommended in this problem as those models achieved very low performance. In future, proposed model can be implemented on other application domain.

ACKNOWLEDGMENT

I am extremely grateful to my Supervisor and my colleagues whose contributions, suggestions on every aspect made this work successful.

REFERENCE

- [1] J. Herzig, M. S. Scheuer, and D. Konopnicki, "Emotion detection from text via ensemble classification using word embeddings",

- International Conference on Theory of Information Retrieval, pp. 269-272, October 2017.
- [2] S. Fathy, N. E. Haggag, and M. H. Haggag, "A hybrid model for emotion detection from text," *International Journal of Information Retrieval Research (IJIRR)*, vol. 7, pp. 32-48, January 2017.
 - [3] M. H. Su, C. H. Wu, K. Y. Huang and Q. B. Hong, "LSTM-based text emotion recognition using semantic and emotional word vectors", *First Asian Conference on Affective Computing and Intelligent Interaction (ACII Asia)*, Beijing, China, pp. 1-6, May 2018.
 - [4] C. R. Chopade, "Text based emotion recognition: A survey", *International Journal of Science and Research*, vol. 4, pp. 409-414, June 2015.
 - [5] T. T. Sasidhar, B. Premjith and K. P. Soman, "Hinglish (Hindi+English) code-mixed social media text," in *Procedia Computer Science*, vol. 171, pp. 1346-1352, 2020.
 - [6] N. Shelke, S. Chaudhury, S. Chakrabarti, S. L. Bangare, G. Yogapriya and P. Pandey, "An efficient way of text- based emotion analysis from social media using LRA-DNN", *Neuroscience Informatics*, vol. 2, September 2022.
 - [7] A. Chatterjee, U. Gupta, M. K. Chinnakotla, R. Srikanth, M. Galley and P. Agrawal, "Understanding emotions in text using deep learning and big data", *Computers in Human Behavior*, vol. 93, pp. 309-317, April 2019.
 - [8] T. Parvin and M. M. Hoque, "An ensemble technique to classify multi-class textual emotion", in *Procedia Computer Science*, vol. 193, pp. 72-81, 2021.
 - [9] D. Yasmina, M. Hajar and A. M. Hassan, "Using YouTube comments for text-based emotion recognition", in *Procedia Computer Science*, vol. 83, pp. 292-299, 2016.
 - [10] S. K. Bharti, S. Varadhaganapathy, R. K. Gupta, P. K. Shukla, M. Bouye, S. K. Hingaa and A. Mahmoud, "Text-based emotion recognition using deep learning approach", *Computational Intelligence and Neuroscience*, August 2022.
 - [11] A. R. Murthy and K. M. A. Kumar, "A review of different approaches for detecting emotion from text", *IOP Conference Series: Materials Science and Engineering*, vol. 1110, pp. 012009, March 2021.
 - [12] S. N. Shivhare and P. S. Khethawat, "Emotion detection from text", *International Conference on: Data Mining and Knowledge Management Process (DKMP 2012)*, May 2012.
 - [13] F. A. Acheampong, C. Wenyu and H. N. Mensah, "Text-based emotion detection: advances, challenges, and opportunities", *Wiley Online Library*, vol. 2, May 2020.
 - [14] A. Chatterjee, K. N. Narahari, M. Joshi, and P. Agrawal, "SemEval-2019 Task 3: EmoContext contextual emotion detection in text", In *Proceedings of the 13th International Workshop on Semantic Evaluation*, pp. 39-48, June 2019.
 - [15] S. X. Mashal and K. Asnani, "Emotion intensity detection for social media data", *International Conference on Computing Methodologies and Communication (ICCMC)*, Erode, India, pp. 155-158, July 2017.